Communication is the process of exchange information from one place to another place.

Distortion happens in composite signals which means signals change its form or shape

multiplexing = multiplexing is the process of simutanously transmitting two or more signals over a single communication channel.

Jitter = a problem occur when different packet of data encounter different delay.

AM: The carrier signal is modulated so the amplitude varies with the changing of the amplitude modulating signal. Frequency and phase remain the same

Demodulator: recovering the message signal from the receive modulated carrier

FM: frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of modulating signal

PM: phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of modulating signal

DSL: digital subscriber line, support high speed digital communication over the existing telephone network.

Digital modulation= encoding a digital data into the amplitude phase or frequency of the carrier signal.

A data element is the smallest entity that can represent a piece of information = the bit.
A signal element is the smallest unity of a signal.
data elements are being carried, signal elements are the carriers.

Amplitude Shift Keying = amplitude of the carrier signal is varied to create signal element.

Multi-level digital modulation (M-ary modulation), stimultaneous transmission of multiple bits by using more than two signal elements.

Pulse Code Modulation :

1. sampling 2. quantization 3. encoding